

Register Transfer & Micro-Operations

Q. What are Micro-Operations Discuss the four main types: Register Transfer, Arithmetic, Logic, and Shift Micro-Operations with examples.

[DEF] Definition to Remember

A Micro-Operation is the most basic, fundamental, atomic operation performed on data stored in CPU registers during one clock pulse. Every high-level machine instruction is broken down into a sequence of these elementary micro-operations.

1. Types of Micro-Operations

There are four primary categories of micro-operations based on their function.

A. Register Transfer Micro-Operations

Transfe

B. Arithmetic Micro-Operations

Perform

C. Logic Micro-Operations

Perform

D. Shift Micro-Operations

Moves

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Micro-Operation: The smallest atomic operation executed in one clock pulse.
2. Four types: Register Transfer, Arithmetic, Logic, Shift.
3. Register Transfer: Moves data without changing it. Example: $R2 \leftarrow R1$.
4. Arithmetic: Basic math (Add, Sub, Inc). Example: $R3 \leftarrow R1 + R2$.
5. Logic: Bitwise operations (AND, OR, XOR) used for masking. Example: $R1 \leftarrow R1 \wedge R2$.
6. Shift: Moves bits left/right for alignment or fast math.
7. Shift types: Logical (fills with 0), Circular (rotates bits), Arithmetic (preserves sign bit).

[RECALL] Quick Recall

Micro-Ops (1 clock pulse) -> 4 Types: Transfer ($R2 \leftarrow R1$) -> Arithmetic ($R1 + R2$, +1) -> Logic (AND/XOR, bit masking) -> Shift (Logical, Circular, Arithmetic for signed math)